Fractions and equal parts

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. 3×5

6. 2×6

7. $60 \div 5$

8. Find both $4 \div \frac{1}{5}$ and $\frac{1}{5} \div 4$.

9. Find $3 \times \frac{2}{3}$.

10. $\frac{2}{3} \times \frac{2}{5}$

11. How many $\frac{1}{3}$-meter pieces fit in 2 meters?

12. Share $\frac{1}{3}$ liter equally among 2 cups. How much per cup?

13. Eli says $3 \div \frac{1}{4} = \frac{3}{4}$, because division makes numbers smaller. Show why this fails.

14. 5 bags each hold $\frac{3}{6}$ kilogram. Find the total mass.

15. Find $\frac{5}{6}$ of $\frac{4}{8}$. Sketch equal parts.

16. Find $\frac{1}{5}$ of $1\ \frac{1}{4}$ liters.

17. A 7-meter cord is cut into $\frac{1}{8}$-meter pieces. How many pieces?

18. Share $\frac{1}{3}$ liter equally among 3 cups. How much per cup?

CHALLENGE / Explain your plan and check the result

C1. A baker has 3 meters of ribbon. Four packages each need $\frac{5}{8}$ meter. Is there enough ribbon? Find the amount left or missing.

C2. A $\frac{1}{3}$-liter sample is shared equally among 4 cups. Another sample gives each cup $\frac{1}{10}$ liter. Which share is larger, and by how much?